

FIM 590 - HOMEWORK 1

ANDREW LYS

Question 1. Write an HJB equation for

$$\inf_{\pi} \mathbb{E} \left[\frac{\gamma}{2} \int_0^T \pi_s^2 + X_T^2 \right]$$

where

$$dX_t = -\pi_t dt + \sigma dW_t$$

with $X_0 = x \neq 0$ and $\gamma > 0$

Solve the HJB using the ansatz

$$H(t, x) = a(t)x^2 + b(t)$$

Solution. We have

$$H(t, x) = \sup_{\pi} \mathbb{E}^x \left[-\frac{\gamma}{2} \int_t^T \pi_s^2 - X_T^2 \right]$$

We have that the infinitesimal generator of X is

$$\mathcal{L}^{\pi} = \frac{1}{2} \sigma^2 \partial_x^2 - \pi_t \partial_x$$

Then we have:

$$\begin{cases} \frac{\partial H}{\partial t} + \sup_{\pi} (\mathcal{L}^{\pi} H - \frac{\gamma}{2} \pi_t^2) = 0 \\ H(T, x) = -x^2 \end{cases}$$

$$\frac{\partial H}{\partial t} + \sup_{\pi} \left(\frac{1}{2} \sigma^2 \partial_x^2 H - \pi_t \partial_x H - \frac{\gamma}{2} \pi_t^2 \right) = 0$$

The supremum is attained at

$$\pi_t = -\frac{1}{\gamma} \partial_x H$$

so we have

$$\begin{aligned} \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \partial_x^2 H + \frac{1}{\gamma} (\partial_x H)^2 - \frac{1}{2\gamma} (\partial_x H)^2 &= 0 \\ \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \partial_x^2 H + \frac{1}{2\gamma} (\partial_x H)^2 &= 0 \end{aligned}$$

Plugging in the ansatz, we have:

$$a'(t)x^2 + b'(t) + \sigma^2 a(t) + \frac{2}{\gamma} a(t)^2 x^2 = 0$$

which gives us the following ODEs:

$$\begin{cases} a'(t) + \frac{2}{\gamma} a(t)^2 = 0 \\ b'(t) + \sigma^2 a(t) = 0 \\ a(T) = -1, b(T) = 0 \end{cases}$$

We can solve the first ODE to get:

$$a(t) = -\frac{\gamma}{2(T-t) + \gamma}$$

and then solve the second ODE to get:

$$b(t) = -\frac{\sigma^2 \gamma}{2} \log \left(\frac{2(T-t)}{\gamma} + 1 \right)$$

and we have that the optimal control is:

$$\pi_t = \frac{1}{T-t + \gamma/2} X_t$$

to summarize:

$$\begin{cases} H(t, x) = -\frac{\gamma}{2(T-t) + \gamma} x^2 - \frac{\sigma^2 \gamma}{2} \log \left(\frac{2(T-t)}{\gamma} + 1 \right) \\ \pi_t = \frac{1}{T-t + \gamma/2} X_t \end{cases}$$

□

Question 2. Consider a risky asset price:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

for a portfolio with self-financing condition:

$$\pi X_t^\pi = X_t^\pi \pi_t \frac{dS_t}{S_t} + r(1 - \pi_t) X_t^\pi dt - c_t dt$$

$c_t \geq 0$.

Find the optimal consumption and the optimal value for

$$\sup_{\pi} \mathbb{E} \int_0^\infty e^{-\rho t} \log(c_t) dt$$

where $\rho > 0$

Question 3. Prove that Brownian motion has infinite total variation almost surely.

Solution. We prove a stronger statement: Let M be a continuous local martingale. If M is almost surely non-constant, then M has infinite total variation almost surely.

Let $V_M(t, \omega)$ be the total variation of M on $[0, t]$. We assume for contradiction that $V_M(t) < \infty$ with positive probability. By stopping on the first time that $|M| > n$, we can assume that M is bounded. We then define

$$T_0^n := \inf\{t > 0 : |M_t| > 2^{-n}\}, T_k^n := \inf\{t > T_{k-1}^n : |M_t - M(T_{k-1}^n)| > 2^{-n}\},$$

and $t_k^n := T_k^n \wedge t$. Then, we have that

$$[M]_t^n := \sum_{k \geq 1} (M(t_k^n) - M(t_{k-1}^n))^2 \rightarrow [M]_t$$

uniformly a.s. This is true by Rogers and Williams IV.30.1. (I don't want to rewrite the proof here) However, we also have that

$$[M]_t^n \leq 2^{-n} \sum_{k \geq 1} |M(t_k^n) - M(t_{k-1}^n)| \leq 2^{-n} V_M(t)$$

So, since with positive probability, we have that $V_M(t) < \infty$, then with positive probability, we have $[M]_t = 0$ for $t > 0$. But, since

$$M^2 - [M]$$

is a martingale, stopping this martingale at

$$S := \inf\{t > 0 : [M]_t > 0\}$$

and optionally stopping, we have

$$\mathbb{E}[M_{S \wedge t}^2] = \mathbb{E}[[M]_{S \wedge t}] = 0$$

so that $M_{S \wedge t} = 0$, and since $S > 0$ with positive probability, We have that M is constant with positive probability, which is a contradiction.

Since Brownian Motion is a continuous local martingale, and is almost surely non-constant, we have that Brownian motion has infinite total variation almost surely. $\square$